

BENFORD-CENTRED FINANCIAL INTEGRITY CORE

Complete implementation guidance for the technical team

Twelve protection layers + complementary mathematical, graph, machine-learning and entity-level models

Design objective

Build a core that is difficult to misuse: complete data, explicit suitability, deterministic calculations, independent corroboration, source traceability, transparent prioritisation and final human judgement.

Document item	Value
Version	1.2 - Complete mathematical-method coverage
Date	21 August 2026
Status	Shareable implementation guidance
Source baseline	Ledger Integrity & Audit Intelligence Platform BRD v1.0
Primary audience	Technology lead, data/analytics team, audit methodology lead and QA team

Important limitation

No combination of Benford, mathematics, graph analytics or machine learning can make fraud detection literally foolproof. The correct target is a robust, fail-closed, explainable and reproducible decision-support engine. The platform flags risks; authorised professionals decide.

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How to use this document

This is not a code-level specification. It defines what the integrity core should contain, what each method contributes, the minimum outputs and the safeguards that must remain intact. The technical team can choose the implementation architecture.

1. Purpose and product boundaries

Benford remains the statistical foundation of the platform, but Benford alone cannot detect every suspicious amount, timing pattern, relationship, reconciliation difference or company-level reporting signal. The integrity core should therefore combine several independent method families and bring their results into one explainable investigation case.

Principle	Required behaviour
Data before analytics	Do not run formal analysis until source files, mappings, balances, control totals and source lineage are known.
Suitability before Benford	Reject or caution identifiers, fixed-price sets, small populations, narrow ranges and incomplete populations.
Deterministic core	Benford, reconciliation, rule checks and Review Priority Score calculations are versioned and reproducible.
Independent corroboration	Benford receives limited weight by itself. Priority rises when different method families point to the same records.
Exact traceability	Every result links to the original source file, row, transaction, calculation and methodology version.
No black-box verdict	An anomaly score or model output is a candidate for review, not proof of fraud or misstatement.
Human judgement	Auditors accept, reject, explain, escalate and sign off every professional conclusion.

2. The 12 protection layers

The twelve layers form one control chain. A later layer cannot repair a failed earlier layer. For example, an unsupervised model cannot make an incomplete or unsuitable population reliable.

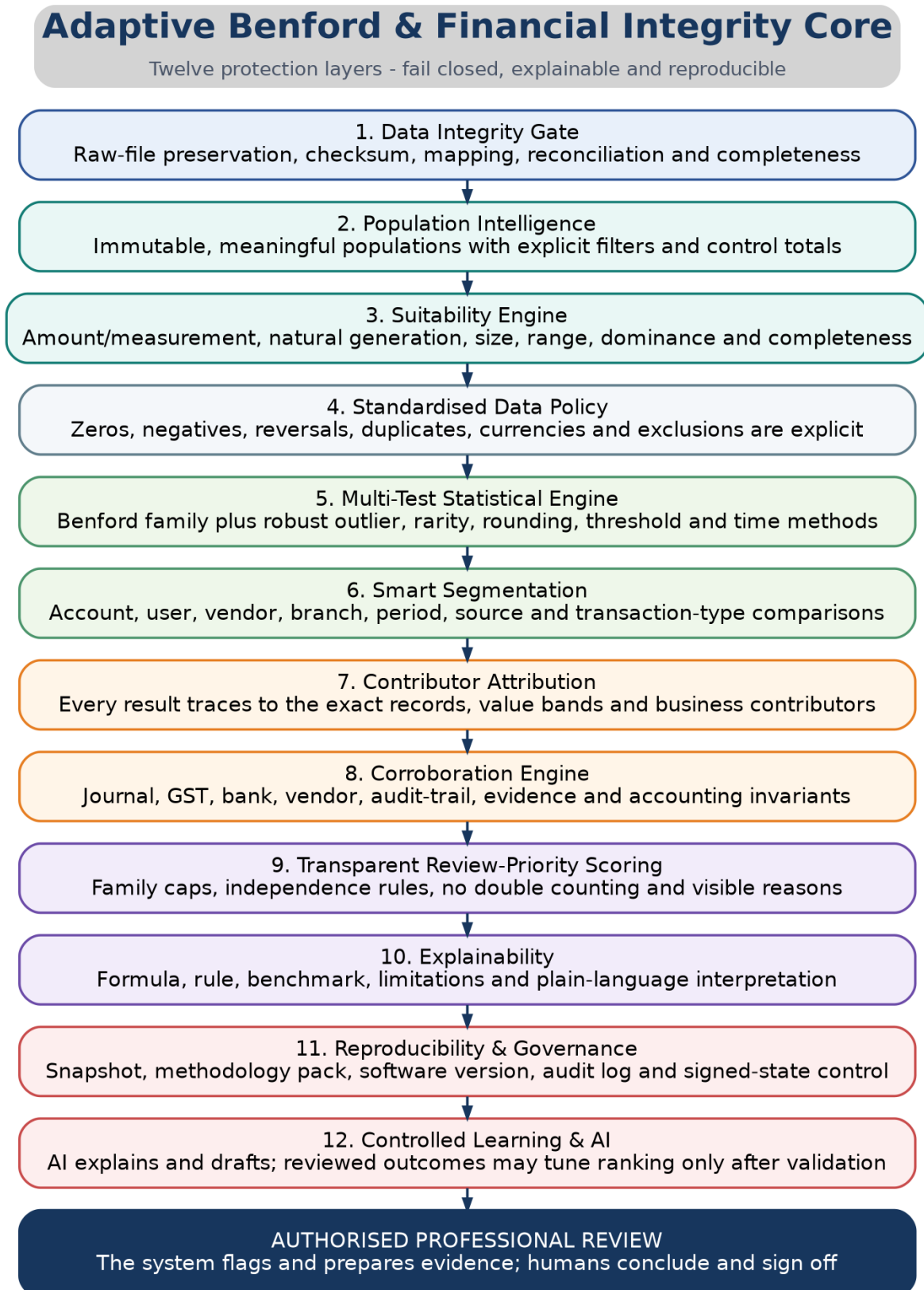


Figure 1. Twelve protection layers for the Benford-centred integrity core.

Layer	What it protects	Minimum expected outcome
1. Data Integrity Gate	Prevents analysis on incomplete, altered or unreconciled data.	Trusted snapshot, checksum, field-quality report and control-total status.
2. Population Intelligence	Prevents blind analysis of the entire ledger or unrelated records.	Saved population definition with entity, period, field, filters, count and value.
3. Suitability Engine	Prevents false confidence from unsuitable Benford populations.	Suitable, Suitable with caution or Not suitable, with visible reasons.
4. Standardised Data Policy	Prevents hidden choices about zeros, negatives, reversals and duplicates.	Explicit inclusion/exclusion register; no source row silently removed.
5. Multi-Test Statistical Engine	Prevents dependence on one digit chart.	Benford plus outlier, rarity, threshold, time and supporting digital tests.
6. Smart Segmentation	Prevents whole-company averages from hiding local problems.	Account, user, vendor, branch, month and type comparisons.
7. Contributor Attribution	Prevents an unexplained population-level finding.	Exact contributing transactions, people, accounts, vendors and value bands.
8. Corroboration Engine	Prevents Benford from standing alone.	Journal, GST, bank, vendor, audit-trail, evidence and invariant signals linked.
9. Transparent Review Priority	Prevents hidden or inflated scores.	Visible family scores, caps, reasons, exposure and override history.
10. Explainability	Prevents technical outputs that reviewers cannot defend.	Plain-language reason, benchmark, limitations and next procedure.
11. Reproducibility and Governance	Prevents results changing silently over time.	Snapshot, methodology pack, software version, execution time and audit log.
12. Controlled Learning and AI	Prevents AI from altering facts or replacing judgement.	AI suggestion with sources, uncertainty and human Accept/Edit/Reject.

3. Mathematical method families

Each method answers a different question. Robustness comes from independent methods agreeing on the same transaction group, not from adding more versions of the same signal.

Method family	Question answered	Recommended delivery
Benford and digital tests	Does the selected population have an unusual digit or repetition pattern?	MVP core
Robust peer-group outliers	Which individual amount is unusual compared with similar transactions?	MVP core
Frequency and rarity	Which user, account pair, vendor, time or transaction combination is rare?	MVP core
Threshold and split detection	Are values shaped just below approval, materiality or policy limits?	MVP core
Time-series and change detection	Did volume, value or behaviour change suddenly or gradually?	Simple baseline in MVP; advanced Phase 1
Reconciliation and invariants	Do the books agree mathematically with control totals and independent records?	Mandatory foundation
Relationship and graph analytics	How are vendors, users, accounts, invoices and payments connected?	Basic deterministic checks in MVP; graphs Phase 1/2

Method family	Question answered	Recommended delivery
Unsupervised anomaly detection	Which multi-dimensional records look unusual even when no predefined rule catches them?	Phase 2/3
Financial-statement models	Does the company-year show manipulation or reporting-risk indicators?	Phase 2/3 specialised layer

3.1 Benford and supporting digital methods

Method	What it contributes	Important treatment
First digit, second digit, first two digits	Population-level conformity and detailed contributor drill-down.	Run only after suitability approval.
Benford Mean Absolute Deviation (distribution MAD)	Measures the overall distance between the observed and expected digit distributions.	Population-level conformity measure; label it separately from Median Absolute Deviation.
Last digit and round numbers	Human preference for 0, 5, 00, 000, 50, 99 or firm-defined endings.	Supporting digital test; not Benford conformity.
Last-two-digit / terminal-pair analysis	Tests concentration across all endings from 00 to 99, including preferences such as 00, 25, 50, 75, 90, 95 and 99.	Compare manual and system-generated populations separately; supporting digital-forensics test.
Exact duplication and repeated values	Amounts repeated more often than expected.	Group exact and near-duplicate records.
Summation test	Digit groups containing unusually high total monetary value.	Supporting diagnostic; Phase 1.
Second-order Benford	Patterns in differences between sorted values; may reveal construction or extraction issues.	Supporting diagnostic; Phase 1.
Median Absolute Deviation + Modified Z-score	Ranks individual amounts that are unusual within a comparable peer group.	Show peer group, median, MAD, score and fallback when MAD equals zero; use log amounts where appropriate.
Rarity score	Rare account pairs, users, timing or business combinations.	Require minimum support and a visible historical frequency.
Threshold bunching	Concentration immediately below approvals or policy limits.	Show grouped value and suspected avoided threshold.

3.2 Keep the two MAD measures separate

The word MAD is used for two different calculations. The product, methodology pack, API output, screen labels and workpapers must never combine or ambiguously label them.

Measure	Level and purpose	Minimum output
Benford Mean Absolute Deviation	Population-level conformity measure. Formula: average of observed digit proportion - expected Benford proportion across the applicable digit categories. It shows how far the complete digit distribution is from expectation.	Observed and expected proportions, per-category absolute differences, number of categories, Benford MAD, methodology threshold and suitability status.
Median Absolute Deviation	Peer-group dispersion measure. Formula: median(x - median(x)). It creates a robust normal range for transaction amounts and is less distorted by extreme values than mean/standard deviation.	Peer-group definition and size, median, median absolute deviation, transformation used, exclusions and zero-MAD treatment.

Measure	Level and purpose	Minimum output
Modified Z-score	Transaction-level ranking derived from the peer-group median and Median Absolute Deviation. Formula: $0.6745 \times (x - \text{median}) / \text{MAD}$. For skewed amounts, the approved methodology may use $\log(1 + \text{amount})$.	Transaction value, peer group, median, MAD, modified Z-score, direction, threshold applied and source-row drill-down.

Zero-MAD safeguard

When every value in a peer group is identical, Median Absolute Deviation equals zero and a Modified Z-score cannot be calculated normally. The engine must not divide by zero. It should mark the score as not calculable, apply an approved fallback such as a broader peer group or exact-frequency rule, and disclose the treatment.

3.3 Last-two-digit / terminal-pair analysis

This supporting test examines the final two digits of valid monetary amounts from 00 to 99. It helps identify manual preferences, constructed amounts and system conventions that a first-digit Benford test may not reveal.

Step	Expected behaviour
1. Define the population	Use a meaningful and comparable group; separate manual and system-generated transactions wherever source metadata permits.
2. Normalise safely	Apply the approved decimal, currency, sign and rounding policy while preserving the original value and source row.
3. Count terminal pairs	Calculate the frequency and percentage for each ending from 00 through 99.
4. Compare with an approved benchmark	Use the methodology-approved expected pattern, historical baseline or comparable system-generated population; do not assume uniformity without approval.
5. Identify contributors	Drill from concentrated endings such as 00, 25, 50, 75, 90, 95 or 99 to users, accounts, vendors, dates and exact transactions.
6. Corroborate	Combine terminal-pair concentration with manual-entry, period-end, threshold, duplicate, reversal, approval and evidence signals.

Interpretation boundary

A concentrated ending can be caused by legitimate pricing, tax rounding, system configuration or normal business convention. Last-two-digit analysis is a supporting signal, not a Benford conformity result and not evidence of fraud by itself.

4. Time-series and change-detection model

Benford examines the shape of a population. Time-series analysis examines whether business behaviour changed over time. It is especially useful for period-end activity, manual journals, reversals, vendor creation, bank-detail changes, GST mismatches and evidence failures.

Item	Implementation guidance
Purpose	Detect sudden spikes, slow drift, unusual seasonality and structural changes in transaction volume, value or behaviour.
Minimum data	Transaction date/time, amount, account, transaction type, user, vendor/customer, branch, source module, reversal/status fields and comparable historical periods.
Main output	Baseline, actual value, deviation magnitude, change start/end date, contributing records and plain-language explanation.
Practical example	Manual journals average 25 per day but rise to 180 per day in the final three days of March; most were posted by two privileged users and reversed in April.
Recommended phase	Rolling baselines and period-end comparison in MVP; EWMA, CUSUM and formal change points in Phase 1.
Key guardrails	Account for seasonality, business events, system migrations, incomplete dates and changes in population definition.

4.1 Methods to include

Method	Simple meaning	Best use
Prior-period comparison	Compare the current period with the same client and a reasonably comparable earlier period.	Year-on-year or month-on-month review.
Rolling median and MAD	Create a robust normal range that is not distorted by a few extreme days.	Daily/weekly journal count, value, reversals and payments.
Period-end concentration	Measure how much annual or monthly activity occurs in the final 30/15/7 days and immediately after close.	Cut-off, late journals, credit notes and reversals.
EWMA	Give more weight to recent observations so small sustained shifts become visible.	Gradual change in manual journals, override activity or mismatch rate.
CUSUM	Accumulate small deviations from normal until a meaningful drift is detected.	Persistent change that may be missed by one-day thresholds.
Change-point detection	Identify when the underlying behaviour changes abruptly.	System changes, new process, control breakdown or new risk pattern.

4.2 Simple workflow

1. Choose one comparable measure: for example daily manual-journal count, daily payment value or weekly reversal rate.
2. Create a clean time series with consistent dates, filters and missing-period treatment.
3. Build the normal baseline using prior periods, rolling median/MAD and known seasonality.
4. Run the selected change method and identify the first date and duration of the change.

5. Drill down from the changed period to exact transactions, users, accounts, vendors and branches.
6. Corroborate with Benford, privileged-user, reversal, threshold, GST, bank and evidence signals.
7. Store the baseline period, parameters, result and source transactions for reproducibility.

Result wording

Use: “Manual journal activity increased materially above the approved baseline during the final three reporting days.” Do not use: “Manipulation occurred.”

5. Reconciliation and accounting-invariant engine

Reconciliation is stronger than an anomaly score because it checks a mathematical or accounting fact. The engine should first establish whether the population is complete and then explain every difference between the books and an independent source.

Item	Implementation guidance
Purpose	Identify exact mathematical inconsistencies and unresolved differences across accounting, GST, bank and transaction sources.
Minimum data	General ledger, trial balance, bank statements, sales/purchase registers, GST returns, invoices, payments, credit notes and source control totals.
Main output	Reconciled totals, matched items, classified differences, unexplained count/value, ageing, exposure and sign-off status.
Practical example	The bank ledger shows Rs. 1.02 crore and the bank statement shows Rs. 98 lakh. Rs. 3 lakh is supported by unpresented cheques and Rs. 1 lakh is an unbooked bank charge. The engine explains the full Rs. 4 lakh difference.
Recommended phase	Mandatory foundation for MVP; richer cross-source and continuous reconciliation in later phases.
Key guardrails	Tolerance rules must be approved; suggested matches require review; legal/tax eligibility remains a professional decision.

5.1 Core accounting invariants

Invariant or reconciliation	Expected relationship	Example exception
Journal balance	Total debit equals total credit for each voucher and the population.	Unbalanced voucher or missing line.
Ledger to trial balance	Ledger movement agrees with trial-balance debit/credit movement.	Incomplete ledger extract or unmapped account.
Opening + movement = closing	Opening balance plus net movement equals closing balance.	Broken balance continuity.
Bank statement continuity	Opening balance plus credits minus debits equals closing balance.	Missing or duplicated bank rows.
Bank-to-books	Book balance adjusted for approved reconciling items equals bank balance.	Unexplained book-only or bank-only item.
Invoice arithmetic	Taxable value + tax + charges - discounts equals invoice total.	Calculation or extraction difference.

Invariant or reconciliation	Expected relationship	Example exception
Invoice-payment lifecycle	Invoice less credit notes and approved adjustments equals outstanding/paid value.	Duplicate payment or unexplained residual.
Books-to-GST	Books reconcile to GSTR-1, GSTR-2B and GSTR-3B through classified differences.	Books-only, portal-only, tax or timing mismatch.

5.2 Simple workflow

1. Preserve each source file and validate opening/closing totals and record counts.
2. Define the expected invariant, keys, tolerances, date windows and grouping rules.
3. Perform exact matching first, then approved tolerance and one-to-many/many-to-one matching.
4. Classify every remaining difference into a business reason; do not leave one undifferentiated unmatched list.
5. Block formal sign-off while material unexplained differences remain unless an authorised override is recorded.
6. Link unresolved differences to the same investigation case as related Benford, journal, vendor, bank or evidence signals.

Why this strengthens Benford

A Benford deviation says the number pattern is unusual. A reconciliation difference says a specific amount does not agree with an independent record. When both point to the same transaction group, the case becomes materially stronger.

6. Relationship and graph analytics

Some risks exist in relationships rather than in one transaction. Graph analytics connects vendors, users, bank accounts, invoices, payments, approvals and master-data changes so the reviewer can see the full business path.

Item	Implementation guidance
Purpose	Detect shared identifiers, hidden connections, control conflicts, unusual clusters and concentrated relationships.
Minimum data	Vendor/customer master, user/role master, bank accounts, GSTIN/PAN where authorised, phone/email/address, invoices, payments, journals, approvals and audit-trail events.
Main output	Source-traceable relationship paths, connected groups, concentration measures, newly formed clusters and consolidated cases.
Practical example	Three differently named vendors share one bank account and address, were created by the same user and received payments within 48 hours.
Recommended phase	Exact shared-identifier and basic conflict rules in MVP; richer interactive graph analytics in Phase 1/2.
Key guardrails	Shared identifiers may be legitimate; every edge must link to a source record; authorised whitelisting and privacy controls are mandatory.

6.1 Graph structure

Graph item	Examples
Nodes	Vendor, customer, employee/user, bank account, GSTIN/PAN, phone/email/address, invoice, payment, journal, approval, role and device/IP where authorised.
Edges	Created by, approved by, paid to, owns/uses bank account, shares identifier, linked to invoice, changed by, reversed by, belongs to entity.
Evidence on every edge	Source file, source row, field name, original value, timestamp and methodology version.

6.2 Deterministic graph methods

Method	What it finds	Example
Connected components	Groups of records linked directly or indirectly through one or more shared relationships.	Vendor A and Vendor B connect through the same bank account and user.
Shared-identifier detection	Exact or approved normalised matches on bank account, GSTIN, PAN, phone, email or address.	Four suppliers use one bank account.
Degree / connection count	Entities with unusually many connections.	One user created 70% of new vendors; one bank account is linked to six vendors.
Unusual cycles	Circular or conflicting paths that may indicate process bypass or money returning through related relationships.	User creates vendor, approves invoice and releases payment through a connected account.
Shortest relationship path	The smallest source-supported path explaining how two entities are connected.	Vendor -> bank account -> another vendor -> creating user.
Newly formed clusters	Groups of relationships created in a short recent window.	Three new vendors, one address and one user appear during the final month.
Vendor-user-bank concentration	Concentration of payment value or activity around a small connected group.	Two users and three bank accounts control 65% of high-risk payments.

6.3 Simple workflow

1. Standardise and tokenise identifiers without changing the original source values.
2. Create nodes and source-linked edges only from authorised data.
3. Run deterministic relationship methods and identify clusters, paths, degrees and cycles.
4. Apply approved legitimate-relationship whitelists without deleting the raw relationship.
5. Link related transaction, user, vendor and payment alerts into one investigation case.
6. Display a simple path and timeline that a non-technical reviewer can explain.

7. Unsupervised anomaly-detection models

Unsupervised models can identify multi-dimensional records that do not resemble normal activity, even when no explicit rule captures the pattern. They should be introduced only after the deterministic core is stable and enough clean data exists.

Item	Implementation guidance
Purpose	Rank transactions that look unusual across several dimensions at the same time.
Minimum data	Clean historical transactions plus approved features such as log amount, account pair, user, branch, posting hour, days from close, vendor age, approval delay, reversal lag, source type and narration category.
Main output	Model anomaly score, peer group, top contributing features, nearest comparable records, confidence/limitations and model version.
Practical example	A payment is not unusually large, but it combines a newly created vendor, rare account pair, after-hours posting, privileged user and unusually short approval time.
Recommended phase	Phase 2/3 only; never a dependency for MVP commercial readiness.
Key guardrails	No cross-client training without permission; version training data and features; monitor drift; show explanations; never use the score as a fraud probability.

7.1 Suitable model options

Model	Strength	Main caution	Recommended use
Isolation Forest	Fast and practical for large, mixed-feature datasets; isolates globally unusual records.	Sensitive to feature design and contamination settings.	First unsupervised model to pilot.
Local Outlier Factor	Finds records unusual compared with their local peer group.	Sensitive to scaling, neighbourhood size and sparse categories.	Peer-group anomaly ranking.
One-Class SVM	Learns a boundary around normal records.	Can be computationally expensive and sensitive to parameters.	Smaller, carefully standardised populations.
Autoencoder	Captures complex nonlinear combinations and reconstruction error.	Needs substantial clean data and is harder to explain.	Later research / high-volume clients.
Clustering + distance	Groups similar transactions and ranks those far from a cluster centre or in tiny clusters.	Choice of cluster count and distance may change results.	Explainable exploration and segmentation.

7.2 Required governance

- Use the model only to rank candidates; deterministic facts and source records remain authoritative.
- Record the training window, population, feature list, scaling, parameters, model version and execution time.
- Validate separately by client type, industry, transaction type and volume; one model may not suit all populations.
- Show why a record ranked highly using top features, peer comparison and nearest normal examples.

- Monitor drift and false-positive rates using auditor-reviewed outcomes.
- Retain a non-AI/non-ML deterministic mode for sensitive engagements.

8. Financial-statement manipulation models

These models operate at the company-year level rather than the individual-transaction level. They identify financial reporting patterns that deserve deeper transaction testing. They should remain a separate entity-level layer that sends questions back into the journal, revenue, expense, bank and evidence modules.

Item	Implementation guidance
Purpose	Identify company-level reporting indicators involving accruals, receivables, margins, cash conversion, asset quality and unusual financial-statement movement.
Minimum data	Preferably three or more years of balance sheet, profit and loss, cash flow, notes, sales/receivable/inventory data and comparable industry/peer information where methodologically valid.
Main output	Entity-year red-flag profile, component ratios, trend, limitations and recommended transaction-level procedures.
Practical example	Revenue grows 40%, receivables grow 90%, operating cash flow becomes negative, gross margin improves sharply and large credit notes appear immediately after year end.
Recommended phase	Phase 2/3 specialised module; not part of the initial Benford screening package.
Key guardrails	Models were developed for specific contexts and must be validated for Indian companies, industry, accounting framework and data quality; no model proves manipulation.

8.1 Recommended company-year methods

Model / method	What it examines	How it strengthens the core
Beneish M-Score	Receivable growth, gross margin deterioration, asset quality, sales growth, depreciation, overheads, leverage and total accruals.	Highlights entity-year conditions associated with earnings manipulation risk and directs testing to revenue, assets and accruals.
Dechow F-Score	Accrual quality and financial-performance variables associated with misstatement risk.	Provides a separate company-level misstatement-risk indicator.
Accrual-quality models	Difference between accounting earnings and cash-based performance; abnormal or persistent accruals.	Directs testing to provisions, estimates, receivables, inventory and capitalisation.
Cash-flow versus profit divergence	Profit or EBITDA movement compared with operating cash flow, receivables and working capital.	Flags growth that is not supported by cash conversion.
Margin and ratio anomaly models	Gross margin, expense ratios, debtor days, inventory days, capitalisation ratios and peer/prior-period changes.	Converts broad statement-level anomalies into accounts and periods for detailed review.

Positioning

Entity-level models answer “Which company-year areas deserve deeper review?” Transaction-level methods answer “Which exact records should the auditor inspect?” Both are useful, but they are not interchangeable.

9. Review Priority Score and worked examples

The score is a transparent ordering mechanism for audit work. It is not a Fraud Probability Score. Every score must show the facts, method families, caps and source records that produced it.

9.1 Starting score families from the BRD

Family	Example signals	Illustrative maximum
Reconciliation risk	Books-GST, bank-books, invoice-payment and balance differences	25
Deterministic rule risk	Duplicate, split, backdated, missing approval, post-close	25
Behaviour and access risk	Privileged user, segregation conflict, unusual timing, role change	15
Statistical risk	Benford, frequency, trend, ratio and peer anomalies	10
Relationship risk	Shared bank account, linked vendors, related users or entities	15
Evidence risk	Missing, inconsistent, altered or insufficient support	10

Starting design only

The family caps above come from the BRD as an illustrative structure. The Audit Methodology Committee must approve the final weights, thresholds, interaction rules and priority bands before production use.

9.2 Scoring rules

- Calculate each raw signal deterministically and preserve it even if a reviewer later changes case priority.
- Cap related signals inside one family. First digit, first-two digits, round values and repeated values must not be added as if they were four independent facts.
- Increase priority when independent families point to the same records: for example statistical + behaviour + reconciliation + evidence.
- Consolidate duplicate alerts before scoring so one underlying event becomes one case.
- Display estimated exposure separately from confirmed misstatement and recovered amount.
- Keep data-quality confidence visible. Material incompleteness should block formal scoring rather than simply reduce the score.
- Allow authorised priority override without changing raw method results; store reason, user and timestamp.

9.3 Simple score structure

Conceptual formula

Review Priority Score = capped Reconciliation + capped Deterministic + capped Behaviour/Access + capped Statistical + capped Relationship + capped Evidence. The total is limited to 0-100. Cross-family corroboration is reflected through approved interaction rules without exceeding the family caps.

Starting band	Suggested operational meaning	Required action
0-14	Routine / low priority	Retain result; review only if selected by sampling or professional judgement.
15-29	Moderate priority	Targeted transaction review and basic explanation.
30-49	High priority	Prompt investigation and supporting evidence.
50-69	Very high priority	Manager review, consolidated evidence request and escalation consideration.
70-100	Immediate partner attention	Urgent review and controlled escalation; still not a fraud verdict.

These bands are proposed implementation guidance, not a final BRD requirement. They must be validated with pilot outcomes and approved methodology.

9.4 Worked scoring examples

Case	Family scores (illustrative)	Total / treatment
A. Benford deviation only	Statistical 7/10; all other families 0.	7 - Low. Inspect a targeted sample; do not escalate automatically.
B. Benford + round values + threshold bunching	Statistical 8/10 (Benford and round values capped together); Deterministic 12/25 (threshold group).	20 - Moderate. Review grouped entries and approval logic.
C. Benford + privileged user + post-close entry	Statistical 7/10; Deterministic 13/25; Behaviour/Access 12/15.	32 - High. Request approval and supporting calculation.
D. Benford + quick reversal + GST mismatch + missing evidence	Statistical 8/10; Deterministic 14/25; Reconciliation 20/25; Evidence 9/10.	51 - Very high. Create one consolidated case and escalate for manager review.
E. Shared bank cluster + duplicate payments + same user create/approve	Relationship 15/15; Deterministic 18/25; Behaviour/Access 13/15; Reconciliation 12/25; Evidence 8/10.	66 - Very high. Investigate vendor relationships, approval conflict, bank support and onboarding evidence.

9.5 Full case example

Item	Illustrative result
Population	9,240 manual journal lines for one financial year.
Benford result	First-two-digit groups 48 and 49 materially exceed expectation; Statistical family = 8/10.
Time result	67% of the contributing entries were posted in the final seven days; Deterministic/period-end contribution included.
Behaviour result	71% were posted by two privileged users; Behaviour/Access = 12/15.
Reversal result	64% were reversed within seven days; Deterministic family = 15/25 after family cap.
Evidence result	Supporting schedules are missing for the highest-value group; Evidence = 8/10.
Review Priority Score	43/100 - High priority under the proposed starting bands.
System wording	“Unusual numerical pattern with period-end, privileged-user, reversal and evidence signals.”
Prohibited wording	“Fraud detected” or “fabricated journal confirmed.”

10. How all methods connect

Every method should publish a source-linked signal into one common case-consolidation flow. The system must preserve individual method results while avoiding duplicate cases and double counting.

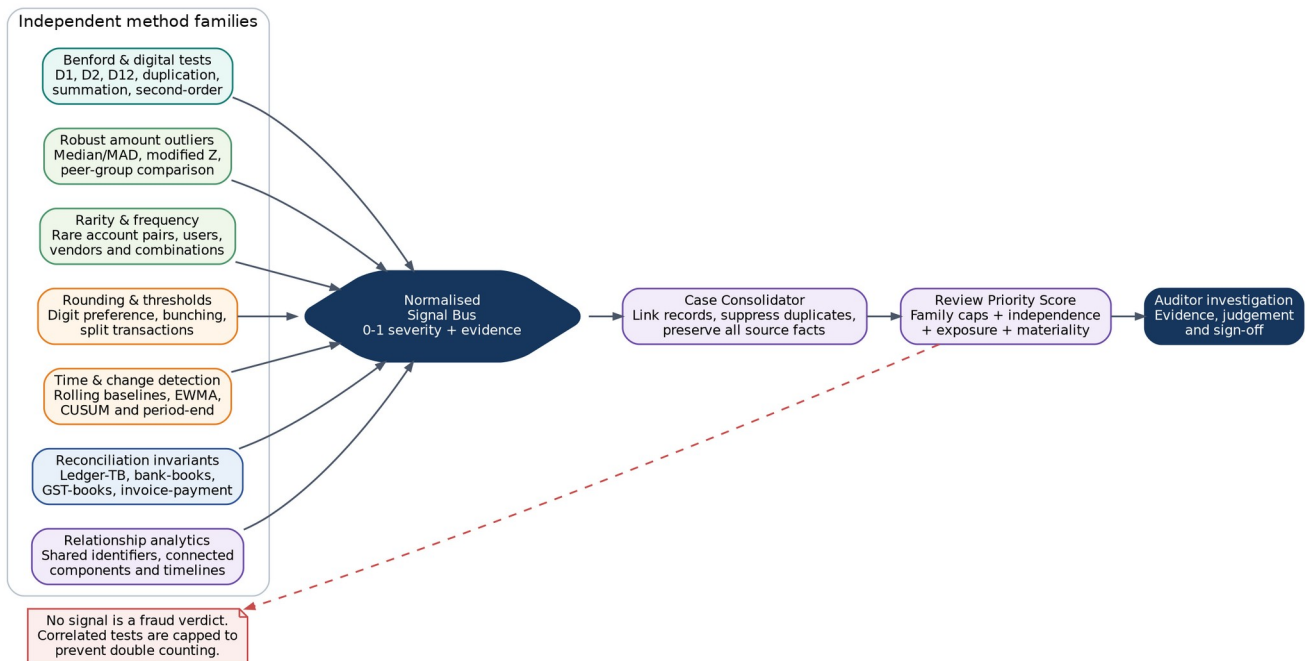


Figure 2. Independent method families feed one consolidated case and transparent Review Priority Score.

Stage	What happens	Output
1. Trusted data	Import, checksum, mapping, completeness and control-total validation.	Immutable source snapshot.
2. Population and suitability	Define a meaningful population and decide whether Benford is appropriate.	Saved population and suitability decision.
3. Deterministic analytics	Run Benford, outlier, rarity, threshold, time, reconciliation and basic relationship methods.	Source-linked signals.
4. Advanced analytics	Where approved, add graph, unsupervised and company-year indicators.	Additional candidate signals with model limitations.
5. Case consolidation	Link signals by transaction, vendor, user, account, time, bank and document relationships.	One case per underlying event.
6. Review Priority	Apply family caps and visible interaction rules.	Explainable 0-100 priority score.
7. Evidence and judgement	Request documents, record explanations and perform professional review.	Approved conclusion and workpaper.

11. Delivery roadmap: MVP to later phases

Phase	Include	Do not make dependent on
MVP - deterministic foundation	Data gates, population/suitability, Benford D1/D2/D12 with Benford distribution MAD, last-digit and last-two-digit tests, round/repeated values, peer-group Median Absolute Deviation and Modified Z-score, rarity, threshold/split, period-end baseline, ledger-TB and activated-source reconciliations, shared identifier rules, contributor drill-down, case consolidation and Review Priority Score.	Unsupervised ML, autoencoders or company-level manipulation scores.
Phase 1 - richer detection	Summation and second-order Benford, EWMA/CUSUM, formal change points, richer graph paths/clusters/concentration, prior-period comparability automation.	Opaque model conclusions.
Phase 2 - advanced intelligence	Isolation Forest or selected unsupervised model, interactive relationship graph, industry benchmarks, continuous controls monitoring, company-year manipulation indicators.	Automatic fraud labels or autonomous closure.
Phase 3 - calibrated learning	Use auditor-reviewed outcomes to tune ranking, monitor drift and validate model performance.	Changing historical deterministic results or signed work.

MVP discipline

Do not delay the commercial core while waiting for machine learning. A deterministic engine with strong data gates, suitability, reconciliation, contributor attribution and transparent scoring is more defensible than an early opaque anomaly model.

12. AI role and safeguards

AI sits around the integrity core, not inside the mathematical truth layer. It receives structured, approved facts and may explain or draft. It cannot change the data, formulas, score or professional conclusion.

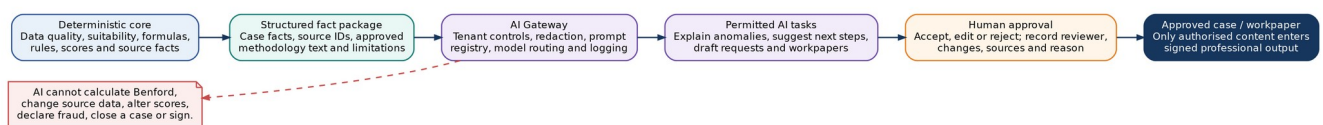


Figure 3. AI assistance is separated from deterministic calculation and requires human approval.

AI may assist with	AI must not do autonomously
Explain Benford, time-series, graph and model signals in plain language.	Calculate or alter Benford, reconciliation, invariant or Review Priority results.
Suggest investigation steps and likely supporting documents.	Change source records, thresholds, methodology or model versions.
Classify narrations and documents with visible confidence.	Declare fraud, tax evasion or misstatement.

AI may assist with	AI must not do autonomously
Draft evidence requests and source-grounded workpaper sections.	Decide GST eligibility, close a case or sign a workpaper.
Summarise management explanations while preserving contrary evidence.	Hide uncertainty, unsupported assumptions or source limitations.

- Label every output as an AI suggestion.
- Show the source records and methodology text used to produce the suggestion.
- Require Accept, Edit or Reject and log the reviewer decision.
- Store model/provider version, prompt template, output and timestamp under the approved retention policy.
- Do not use client data to train shared models without explicit contractual permission.
- Support a non-AI deterministic operating mode.

13. Minimum acceptance checklist and test packs

The core should not be considered ready for pilot use until the following conditions are demonstrated using known datasets and independently recomputed results.

No.	Pass condition
1	Incomplete or materially unreconciled data blocks formal analysis.
2	Identifiers, codes and sequential numbers are rejected as unsuitable for Benford.
3	Fixed-price, narrow-range and small populations receive caution or not-suitable treatment.
4	Counts and values before and after exclusions are visible; no source row is silently removed.
5	Benford Mean Absolute Deviation and transaction-level Median Absolute Deviation are calculated, labelled and stored as separate measures.
6	Last-two-digit analysis displays the 00-99 terminal-pair distribution, benchmark used and exact contributing transactions.
7	Modified Z-score output shows the peer group, median, Median Absolute Deviation, transformation, score and approved zero-MAD fallback.
8	The same source snapshot and methodology version reproduce the same result.
9	Every statistic, graph path and model candidate drills down to exact source records.
10	Time-series results expose baseline, actual result, change window and contributors.
11	Every accounting invariant shows the formula, sources, tolerance and unresolved difference.
12	Every graph edge is source-linked and legitimate shared relationships can be approved/whitelisted.
13	Unsupervised outputs show model version, feature set, peer comparison and limitations.
14	Company-year indicators remain separate from transaction-level findings and never state manipulation as fact.

No.	Pass condition
15	Correlated signals are capped and do not double count the same underlying issue.
16	Review Priority Score shows family scores, caps, reasons, exposure and override history.
17	Related journal, GST, bank, vendor, timing, graph and evidence signals form one understandable case.
18	No screen, alert or export states that a mathematical or machine-learning signal proves fraud.
19	AI outputs are source-grounded, labelled and approved before workpaper use.
20	Signed workpapers are locked, versioned and linked to the exact data and methodology used.

13.1 Golden test packs

- Natural Benford-like population; identifier sequence; fixed-price population; small and narrow-range populations.
- Injected first-two-digit concentration with known contributors, zeros, negatives, reversals, duplicates and invalid values; independently recompute Benford distribution MAD.
- Terminal-pair populations with known 00, 25, 50, 75, 90, 95 and 99 concentrations, including separate manual and system-generated benchmarks.
- Known peer-group outliers for Median Absolute Deviation and Modified Z-score, including skewed amounts, log transformation, small peer groups and a MAD-equals-zero fallback case.
- Known rare combinations, round-value concentration and threshold-splitting groups.
- Stable time series, sudden spike, gradual EWMA/CUSUM drift, seasonality and period-end reversal case.
- Balanced and unbalanced journals, ledger-to-TB difference, bank-book difference, GST mismatch and invoice-payment break.
- Vendor graph with shared bank account, high-degree user, unusual cycle, shortest path and newly formed cluster.
- Synthetic unsupervised dataset with known multi-feature anomalies and a normal peer group.
- Company-year cases with revenue/receivable divergence, abnormal accruals, cash-flow weakness and margin anomalies.
- Worked Review Priority cases that prove family caps, case consolidation and no double counting.
- AI prompt-injection and hallucination tests showing that model output cannot alter deterministic facts.

14. Handover and BRD alignment

This guidance defines the product and methodology intent. The technical team may choose the programming languages, services, database, job orchestration, graph store, model framework, cloud platform and UI architecture, provided the protection layers and acceptance conditions remain intact.

Responsibility	Scope
Technical team	Architecture, storage, APIs, orchestration, graph implementation, model serving, observability, security engineering, performance and deployment.
Audit methodology team	Population rules, suitability thresholds, formulas, model acceptance, scoring weights/caps, wording, evidence expectations and overrides.
GST subject-matter team	GST matching, difference categories, tax exposure wording and filing-readiness controls.

Responsibility	Scope
Information security team	Tenant isolation, encryption, access, audit logging, retention, support access and AI data routing.
Audit users	Population relevance, procedures, evidence sufficiency, exception classification, escalation and final sign-off.

14.1 Key BRD alignment

BRD area	Relevant direction
Data foundation - DAT-001 to DAT-006	Preserve files and checksums, reconcile to control totals, report quality issues and retain source row lineage.
Benford - BEN-001 to BEN-015	Suitability before scoring, explicit exclusions, D1/D2/D12, approved deviation statistics including clearly labelled Benford MAD, supporting terminal-digit tests, contributor drill-down, segmentation, corroboration, reproducibility and neutral wording.
Time and trend - REM-006, PET-003	Compare current activity with prior periods, normal seasonality and visible baselines.
Reconciliation - GST, BKR and DAT requirements	Classify differences, preserve source versions, support exact/tolerance/group matching and block unexplained sign-off where required.
Graph - VPA-007 and related requirements	Link vendors, users, bank accounts, invoices and payments; every relationship must trace to a source record.
Risk scoring - RSK-001 to RSK-005	Explain scores, consolidate related alerts, version weights, allow reasoned override and separate exposure from confirmed misstatement.
AI - AI-001 to AI-005	AI suggestions require source links, uncertainty, human approval, approved data handling and deterministic fallback.

Final design position

Benford remains the statistical foundation. The core becomes robust when independent methods - amount, rarity, time, reconciliation, relationships, evidence and later machine-learning/entity-level indicators - converge on the same records, while every fact remains traceable and the authorised professional retains final judgement.